

TD-COS-FH-001 — L=2 Frozen-Prose

Execution Receipt v0.1

Run date: Sep 6, 2026

Frozen source:  # Toroidal Dynamics v0.1

Later comparison record:  # Tiny-system validation v0.1

Disposition

INVARIANTS PASS · EIGHT-SECTOR COVERAGE PASS · MIXING EVIDENCE FAVORABLE BUT INCOMPLETE · PRODUCTION NOT AUTHORIZED · PHYSICAL Q2 NOT RUN.

Execution

- Kernel: TD-COS-FH-001; cosine weights; fixed-reference Z000; $J=1$, $t=1/2$, $h_6=0$.
- Geometry: $2 \times 2 \times 2$ periodic torus with $V=8$, $E=24$, $P=24$ and $B=71$ microticks per sweep.
- Eight chains: $q=000$ through 111; proposal seeds 64000–64007; acceptance seeds 74000–74007.
- Per chain: 2,000 warmup sweeps and 10,000 retained sweeps; first retained microtick 142,071; last 852,000.
- Completed: 6,816,000 microticks; 80,000 retained measurements; no thinning.
- Outcomes: 1,107,368 accepted nonidentity updates; 4,939,515 rejected nonidentity proposals; 769,117 identity updates.

Invariant evidence

Status: PASS. The sampler executed 1,187,464 online checks with zero failures.

- Integer-current conservation: $\text{div } I=0$.
- Mod-two membrane constraint: $I \bmod 2 + \text{boundary}(M) + \text{Gamma}(q)=0$.
- All transverse cuts agree on signed winding for each axis.
- Sector parity agrees with signed winding: $q_{\alpha}=W_{\alpha} \bmod 2$.
- Cached N_M equals binary membrane occupancy.
- All 123 proposal descriptors preserve the constraints and their tested inverses restore the state.
- Every Metropolis comparison resolved at $K=16$ with one 64-bit prefix; no precision refinement was needed.

Coverage

Every chain visited all eight q sectors after warmup at event level and at retained-sweep resolution. Positive and negative signed winding occurred on every axis. Observed winding ranges: $x=[-6,+6]$, $y=[-6,+6]$, $z=[-5,+6]$. Maximum observed $|l_link|=5$; this was not a cutoff.

- q=000: 10,549 retained states (13.1862%).
- q=001: 10,195 retained states (12.7438%).
- q=010: 10,110 retained states (12.6375%).
- q=011: 9,662 retained states (12.0775%).
- q=100: 10,138 retained states (12.6725%).
- q=101: 9,882 retained states (12.3525%).
- q=110: 9,868 retained states (12.3350%).
- q=111: 9,596 retained states (11.9950%).

Mixing evidence

- x parity: odd fraction 0.4916875; rank-normalized split R-hat 1.0000772; rank bulk ESS 29,471; 16,682 raw $0 \rightarrow 1 \rightarrow 0$ trips.
- y parity: odd fraction 0.4904500; rank-normalized split R-hat 0.9999984; rank bulk ESS 30,620; 16,959 raw $0 \rightarrow 1 \rightarrow 0$ trips.
- z parity: odd fraction 0.4935500; rank-normalized split R-hat 1.0001487; rank bulk ESS 29,468; 16,835 raw $0 \rightarrow 1 \rightarrow 0$ trips.
- Membrane occupancy: rank-normalized split R-hat 1.0002471; rank bulk ESS 29,432.
- Retained-sweep q residence: median 1 sweep, 95th percentile 4 sweeps, maximum 12 sweeps.

The numerical R-hat and ESS components are favorable. The full mixing gate is not established: these round trips are raw rather than decorrelated/effective, and the separately labelled all-zero-start control was not run.

Preserved deterministic mismatch

The later Drive companion already records an L=2 execution. Both records agree exactly on 6,046,883 nonidentity proposals, but they do not agree bit-for-bit: this run accepted 1,107,368 nonidentity updates versus 1,108,720 in the companion (difference -1,352). The largest retained sector-probability point difference is 0.2688 percentage points.

The source Doc names DYNAMICS.json as machine-readable authority, but its native structure contains only literal unlinked relative-path text and Drive has no retrievable file with that exact name. This run is therefore labeled

FROZEN_PROSE_PROFILE_WITH_UNRECOVERED_MACHINE_AUTHORITY. The mismatch is preserved, not harmonized.

Authorization boundary

This is a reproducible L=2 pre-production engineering run. It does not validate warmup, authorize larger-lattice production, establish a physical timescale, or support confinement, deconfinement, or physical Q2.

Before production consideration

- Recover the authoritative DYNAMICS.json or adopt a dated replacement amendment.
- Resolve the deterministic replay mismatch against the preserved companion.
- Run the separately labelled all-zero-start control.
- Establish effective—not merely raw—sector round trips.
- Execute the larger-lattice validation gates under a new frozen profile.

Artifacts

- TD_COS_FH_001_L2_run_results.json — complete aggregate and per-chain receipt.
- TD_COS_FH_001_L2_samples.npz — 80,000 retained measurements and state digests.
- TD_COS_FH_001_L2_verification.json — independent 16/16 structural verification PASS.
- TD_COS_FH_001_L2_companion_comparison.json — exact preserved comparison.
- TD_COS_FH_001_L2_FIRST_RUN_PACKET_v0.1.zip — complete executable packet.

CC0 1.0 Universal. No production authorization. No physical Q2 result.